

Exercise 5.9

$$Q_n = \frac{R_1 + R_2 + \dots + R_n}{n} \Rightarrow \text{computing this requires storing all rewards}$$

$$\text{Incremental update} \Rightarrow Q_{n+1} = Q_n + \frac{1}{n} (R_n - Q_n)$$

Q_n = old estimate R_n = new reward $\frac{1}{n}$ = step size

The same incremental averaging method used for bandits can be applied to Monte Carlo prediction by replacing rewards with returns.

$$\text{The update becomes} \Rightarrow V_{(St)} \leftarrow V_{(St)} + \frac{1}{N(St)} \times [G - V_{(St)}]$$

$N(s)$: number of first visits to state s (single int)

$V(s)$: estimated value of state s (single float)

The original algorithm stores all returns in $\text{Returns}(s)$ and calls $\text{average}(\text{Returns}(s))$. We replace both lines with a counter and an incremental update. This removes the need to store the list entirely, instead the value is updated using only the current estimate and visit count, giving constant memory usage.

$N(St)$ is incremented before the update so that the step size $1/N(St)$ correctly reflects the n -th visit using $\frac{1}{n}$. If incremented after, the first visit would divide by zero.

And because the incremental formula computes the exact same running average as the list method, $V(s)$ still converges to $V_{\pi(s)}$ as the number of first visits goes to infinity.

Input: policy π to be evaluated

Initialize:

$V(s) \in \mathbb{R}$, arbitrarily, for all $s \in S$

$N(s) \leftarrow 0$, for all $s \in S$ $\leftarrow$ replaces $\text{Returns}(s)$

Loop forever (for each episode):

 Generate an episode following π :

$S_0, A_0, R_1, S_1, A_1, R_2, \dots, S_{(T-1)}, A_{(T-1)}, R(T)$

$G \leftarrow 0$

 Loop for each step, $t = T-1, T-2, \dots, 0$:

$G \leftarrow G + R(t+1)$

 if S_t is the first visit in the episode:

$N(S_t) \leftarrow N(S_t) + 1$

$\leftarrow$ increment visit counter

$V(S_t) \leftarrow V(S_t) + (1/N(S_t)) * [G - V(S_t)]$

$\leftarrow$ incremental update